

Week 8, Lecture 1 WOD: Answer Key

How Much Did the Prior Matter?

NRES 746, Instructor Answer Key (not for student distribution)

Each item repeats the handout’s question, then gives the answer, a short rationale, and mistakes to watch for. All three panels are Beta-binomial (the setup from Week 7 Wednesday), so the numbers below are exact. Students don’t see any of them; the point is to read the pictures.

Team	Prior	Survived	Prior mean	Observed	Posterior mean
A	Beta(2, 2)	45 of 60	0.50	0.75	0.73
B	Beta(80, 20)	3 of 5	0.80	0.60	0.79
C	Beta(16, 4)	10 of 20	0.80	0.50	0.65

A Beta(α , β) prior is “worth” about $\alpha + \beta$ animals: 4, 100, and 20 for Teams A, B, and C.

Use this WOD to lead straight into the germination example: the supplier prior there (worth 200 seeds, against 40 seeds of data) puts the germination trial between Teams B and C.

Part 1: Read the panels

1a.

Rank the three teams from most to least influenced by their prior. Give a one-line reason for each.

Answer. B (most), then C, then A (least).

- **B:** the posterior is almost on top of the prior; the data barely moved it.
- **C:** the posterior moved well away from the prior but didn’t leave it behind; it sits between the prior and wherever the data point.
- **A:** the prior is nearly flat, so the posterior’s location and width must come almost entirely from the data.

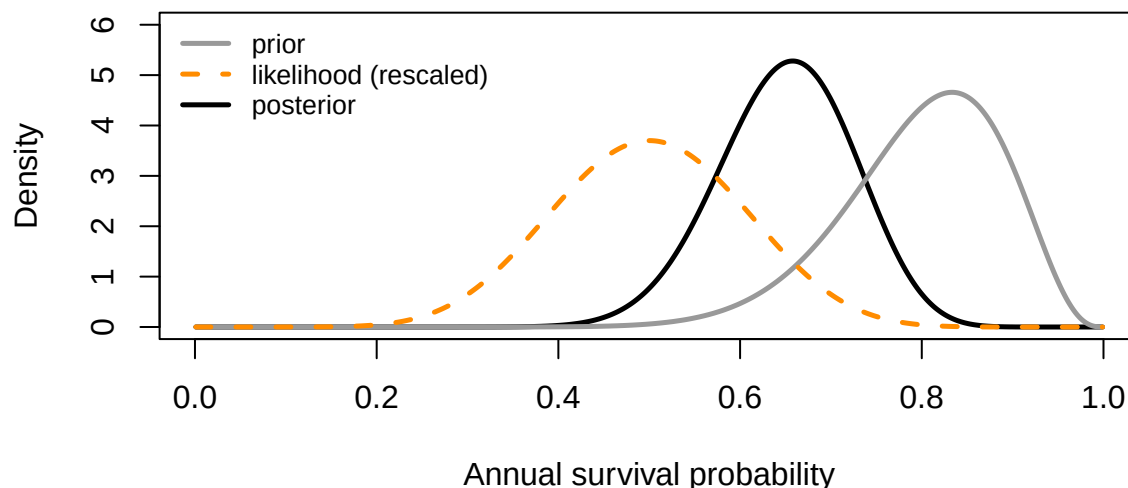
Why. The prior’s influence shows in how far the posterior moved from it and how much narrower it got. Both are tiny for B, moderate for C, and large for A.

Watch for. Ranking by how tall or narrow the posterior is (B looks “most certain” and some students read that as “most data”; see 2a). Calling A’s prior “no prior”: it’s weak, not absent.

1b.

Sketch the shape of Team C’s likelihood on its panel (the height doesn’t matter). Where does it peak, and is it wider or narrower than the prior?

Answer. It peaks at about 0.5, on the far side of the posterior from the prior, and is a little wider than the prior (SD 0.10 vs. 0.09).



Why. The posterior is a compromise between prior and likelihood, so the likelihood has to sit on the opposite side of the posterior from the prior. Here the prior and the data are each worth 20 tortoises, so the posterior mean (0.65) lands halfway between the prior mean (0.80) and the data (0.50). The posterior is narrower than both because it combines two sources of information.

Watch for. Drawing the likelihood between the prior and the posterior (that would pull the posterior *toward* the prior's side). Drawing it narrower than the posterior: the posterior should be the narrowest curve in the panel. Exact position isn't the point; "below 0.65, around 0.5" is full credit.

Part 2: How much data?

2a.

Which team collared the most tortoises, and which collared the fewest? Team B has the narrowest posterior of the three. Does that mean Team B had the most data?

Answer. Team A had the most (60), Team B the fewest (5), Team C in between (20). No: B's posterior is narrow because its prior was narrow, not because of its data.

Why. How much the data contribute shows up in how much the posterior differs from the prior. B's posterior is barely narrower than its prior; A's is far narrower than its (nearly flat) prior. A posterior's width reflects prior and data together, so a narrow posterior alone can't tell you how much data went into it.

Watch for. "B is narrowest, so B had the most data." This is the misconception the question is built to surface. Ask the room: if you only saw Team B's posterior, could you tell whether it came from a strong prior or a big study? (No; you need the prior too. This is why priors must be reported.)